The No-U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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MCMC & Hamiltonian Monte Carlo

- Exact posterior inference is often intractable; MCMC is asymptotically unbiased [Nea93].
- Random-walk & Gibbs [Met+53; GG84] struggle in high-dimensional or correlated spaces.

Hamiltonian Monte Carlo

Auxiliary momentum $r \sim \mathcal{N}(0, I) +$ leapfrog: $p(\theta, r) \propto \exp\{\mathcal{L}(\theta) - \frac{1}{2} r^\top r\}$. Volume-preserving $\Rightarrow$ proposals can be **distant** yet **highly probable**.

Good L requires costly preliminary runs and expert knowledge, a key barrier to adoption.

Step size ε

Too small $\Rightarrow$ wasted computation

Too large $\Rightarrow$ high rejection rate

Leapfrog steps L

Too small $\Rightarrow$ random walk

Too large $\Rightarrow$ trajectory loops back

NUTS addresses both tuning problems simultaneously.

NUTS Methodology

No-U-Turn criterion

Stop when the trajectory doubles back: $(\theta^+ - \theta^-)\cdot r^- < 0$ or $(\theta^+ - \theta^-)\cdot r^+ < 0$.

Recursive doubling

1. Draw direction $v_j \sim \mathrm{Unif}(\{-1, +1\})$
2. Take 2^j leapfrog steps of size $v_j \varepsilon$
3. Grow balanced binary tree; halt on U-turn

Memory $\mathcal{O}(2^j) \to \mathcal{O}(j)$: partition $\mathcal{C} = \mathcal{C}^{\mathrm{old}} \cup \mathcal{C}^{\mathrm{new}}$, propose $w' \in \mathcal{C}^{\mathrm{new}}$, accept with $\min\left(1, \frac{|\mathcal{C}^{\mathrm{new}}|}{|\mathcal{C}^{\mathrm{old}}|}\right)$.

Dual averaging (warm-up only)

Drive acceptance to target δ:

$$x_{t+1} = \mu - \frac{\sqrt{t}}{\gamma(t + t_0)} \sum_{i=1}^{t} H_i,$$

$$\bar{x}_{t+1} = t^{-\kappa} x_{t+1} + (1 - t^{-\kappa}) \bar{x}_t$$

$x = \log \varepsilon$, $\mu = \log(10\varepsilon_0)$

Fixed Hyperparams: $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$

No-U-Turn criterion + dual averaging $\Rightarrow$ NUTS is **entirely tuning-free**.

Evaluating NUTS: The Paper's Benchmarks

The paper tests NUTS on four targets spanning low to very high dimensions, measuring **minimum ESS per gradient evaluation** across all dimensions.

250-D	25-D	302-D	3001-D
Multivariate Normal	Logistic Regression	Hierarchical LR	Stochastic Volatility

NUTS matches or exceeds best-tuned HMC on **all four benchmarks** without any trajectory-length tuning, up to 2× on the Gaussian and 1.5× on the stochastic volatility model.

But all four targets are **unimodal or mildly structured**. How does NUTS fare on posteriors with genuinely complex geometry?

Beyond the Paper: Three Stress Tests

Multimodality

BNN, 191-D

$5! \times 5! =$

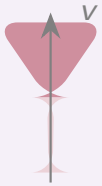
14400 modes

Dense Correlations

GP Classif., 150-D

dense 150×150 cov.

Varying Geometry



Neal's Funnel, 10-D

spatially varying σ

Results

- **BNN:** NUTS explores wider weight space but lower ESS/grad than well-tuned HMC (0.03 vs. 1.57×10^{-4}).
- **GPC:** Modest 40% ESS/grad gain, but NUTS uses $4\times$ more gradient evaluations; advantage is narrow.
- **Funnel:** NUTS recovers $\mathrm{Var}(v) \approx 9$ (truth); HMC's fixed ε fails to explore the neck.

Adaptive trajectory length is critical on complex posteriors; gains is the most dramatic on the Funnel.

Summary & Contributions

Paper assessment

+ Resolves HMC's central practical limitation; conceptually simple yet theoretically rigorous
+ No-U-Turn criterion + recursive doubling preserve detailed balance
+ Dual averaging for step-size adaptation $\Rightarrow$ essentially tuning-free
– Dual-averaging hyperparameters justified empirically; robustness across geometries is open
– Performance guarantees mainly asymptotic; limited range of posterior geometries tested

Our contributions

- **Proofs** of uniform $\mathcal{C}$ distribution and detailed balance of the memory-efficient kernel (only sketched in paper)
- **Reproduced** all 4 benchmarks; NUTS matches or exceeds best-tuned HMC without trajectory-length tuning
- **Stress tests:** adaptive trajectory length is critical on complex posteriors, the gains are most dramatic on Funnel & BNN; advantage narrows on dense GP
- **Conclusion:** confirms NUTS's suitability as a general-purpose inference engine for modern Bayesian models

Bibliography i

[Met+53] Nicholas Metropolis et al. **"Equation of state calculations by fast computing machines".** In: *The journal of chemical physics* 21.6 (1953), pp. 1087–1092.

[GG84] Stuart Geman and Donald Geman. **"Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images".** In: *IEEE Transactions on pattern analysis and machine intelligence* 6 (1984), pp. 721–741.

[Nea93] Radford M Neal. **"Probabilistic inference using Markov chain Monte Carlo methods".** In: (1993).

Appendix: Benchmarks Bayesian & Hierarchical Logistic Regression Efficiency

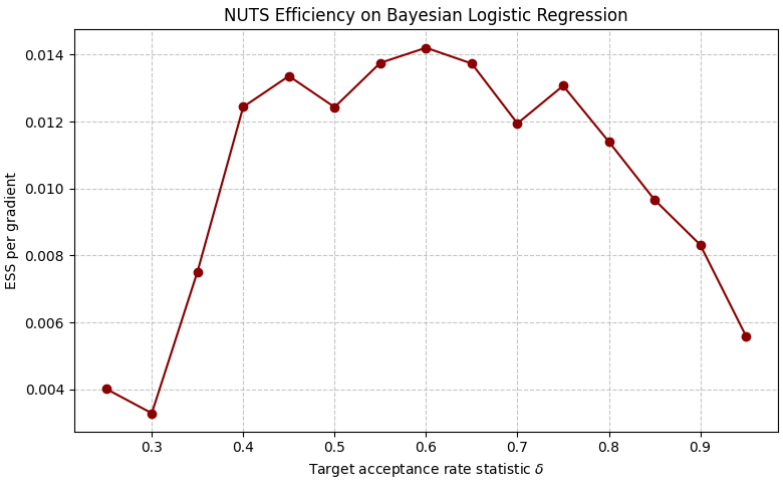


Figure 1: Bayesian Logistic Regression (25-D).

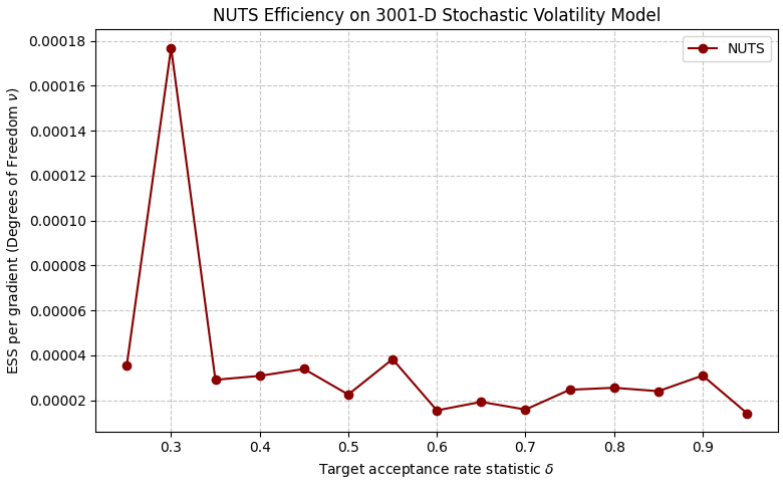


Figure 2: Hierarchical Logistic Regression (302-D).

Appendix: Dual Averaging Convergence

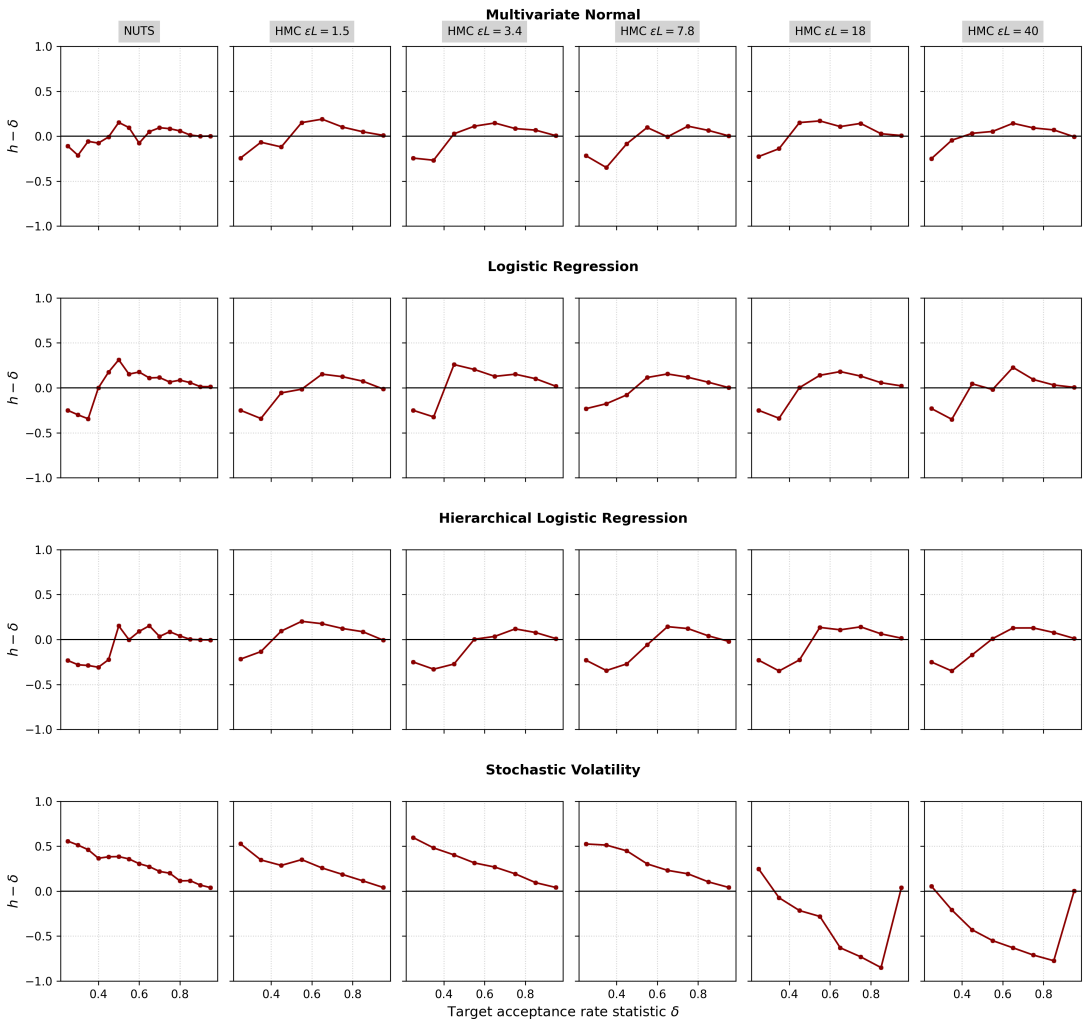


Figure 3: Discrepancy plots ($h - \delta$ across δ) for all four original benchmarks. Dual averaging reliably drives acceptance to target δ within a few hundred warm-up iterations.

Appendix: BNN Trace Plots

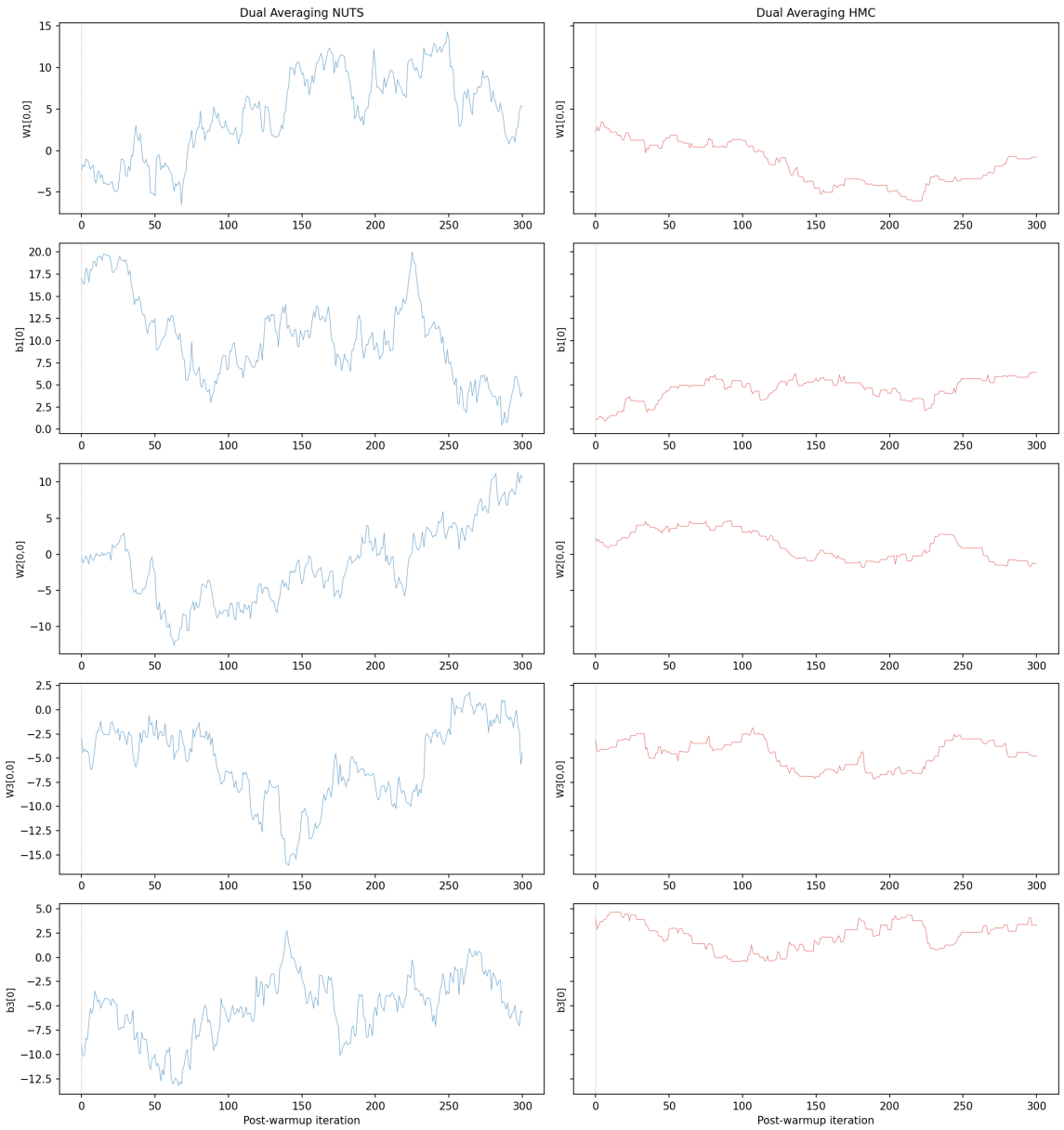


Figure 4: Trace plots: NUTS (blue) covers a broader range of weight values than HMC (red).

Appendix: BNN Adaptation Diagnostics

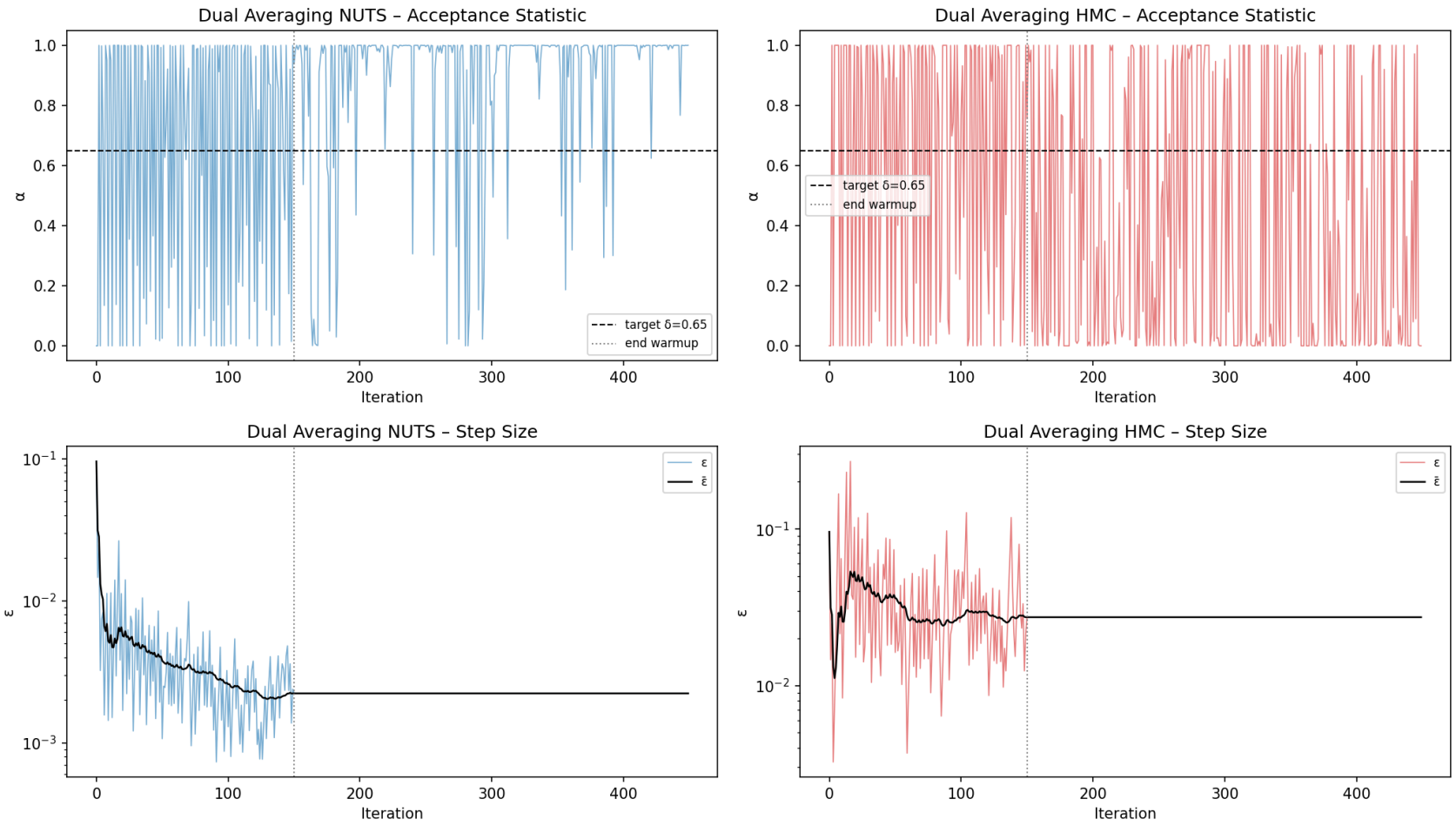


Figure 5: Dual-averaging convergence on the BNN. Step size $\bar{\varepsilon}$ stabilises within the warm-up phase.

BNN: Efficiency

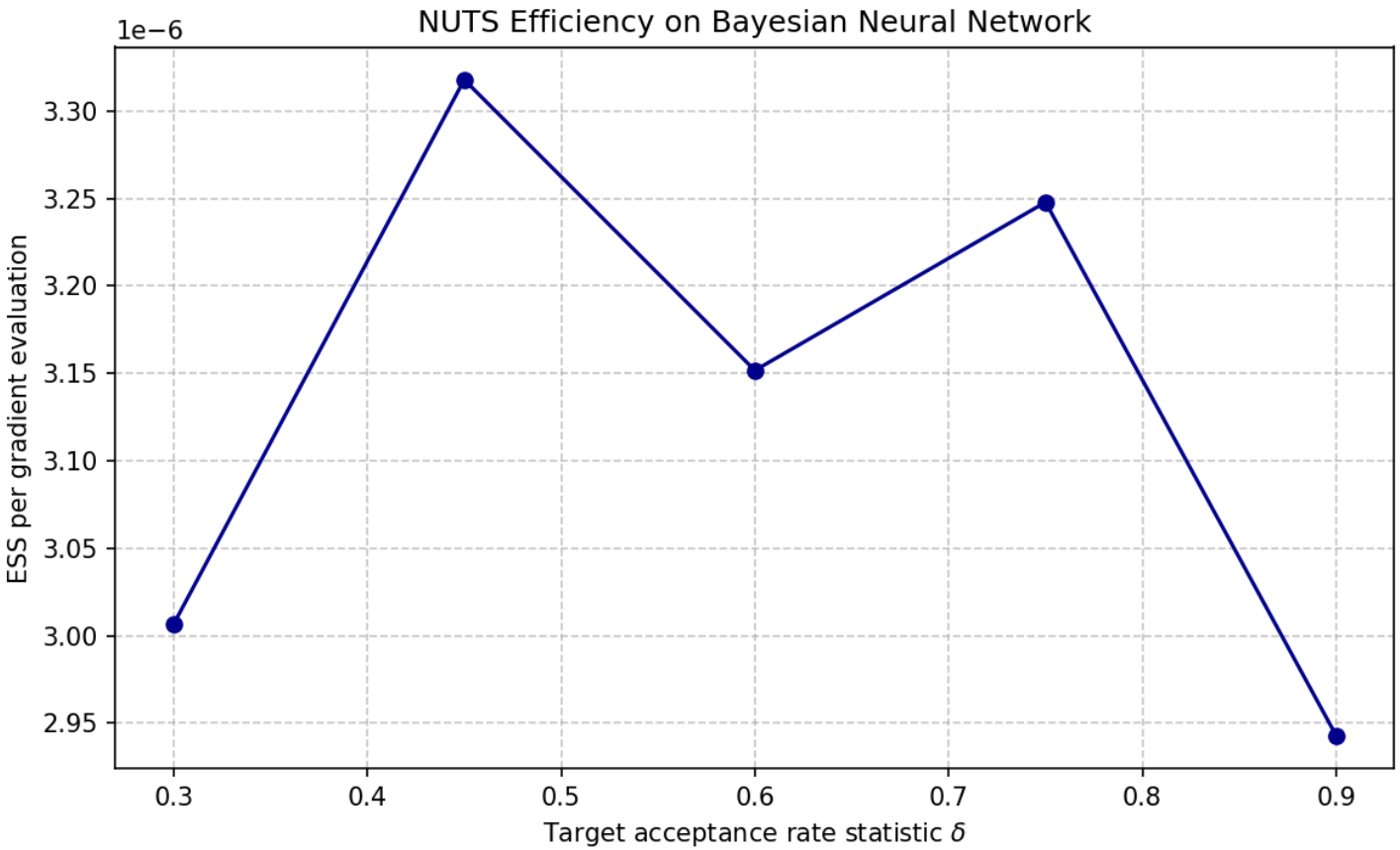


Figure 6: ESS/gradient on the BNN. NUTS consistently achieves higher effective samples per gradient evaluation than HMC across all tested configurations.

Appendix: GPC Kernel Matrix & Posterior Correlations

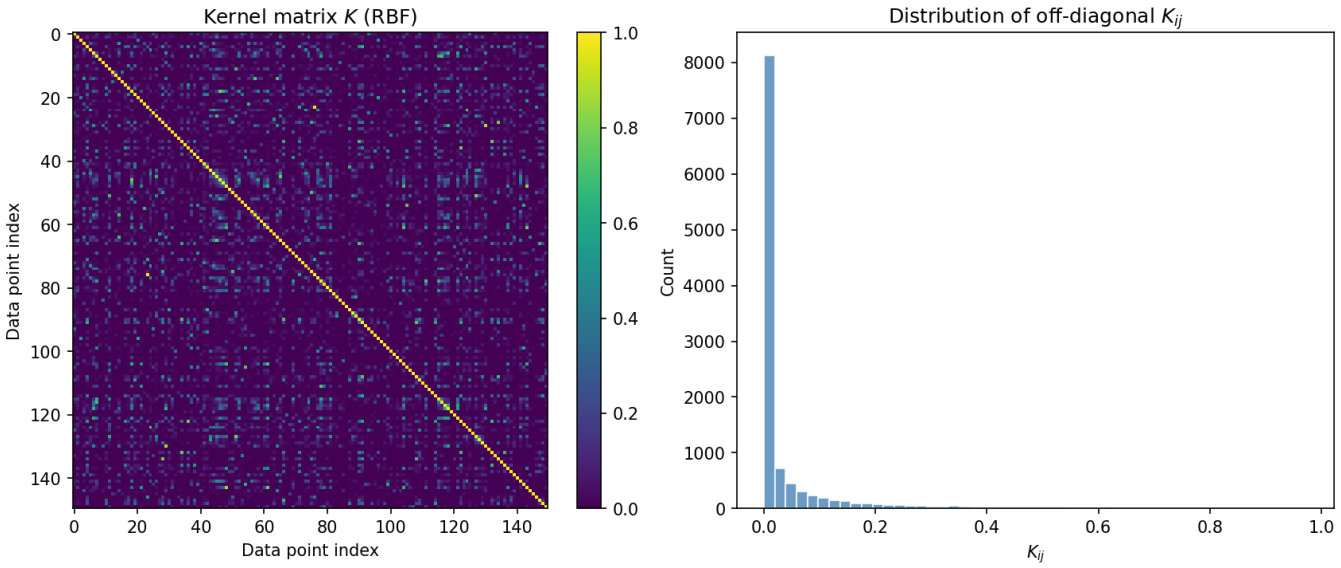


Figure 7: RBF kernel matrix K: dense correlation structure.

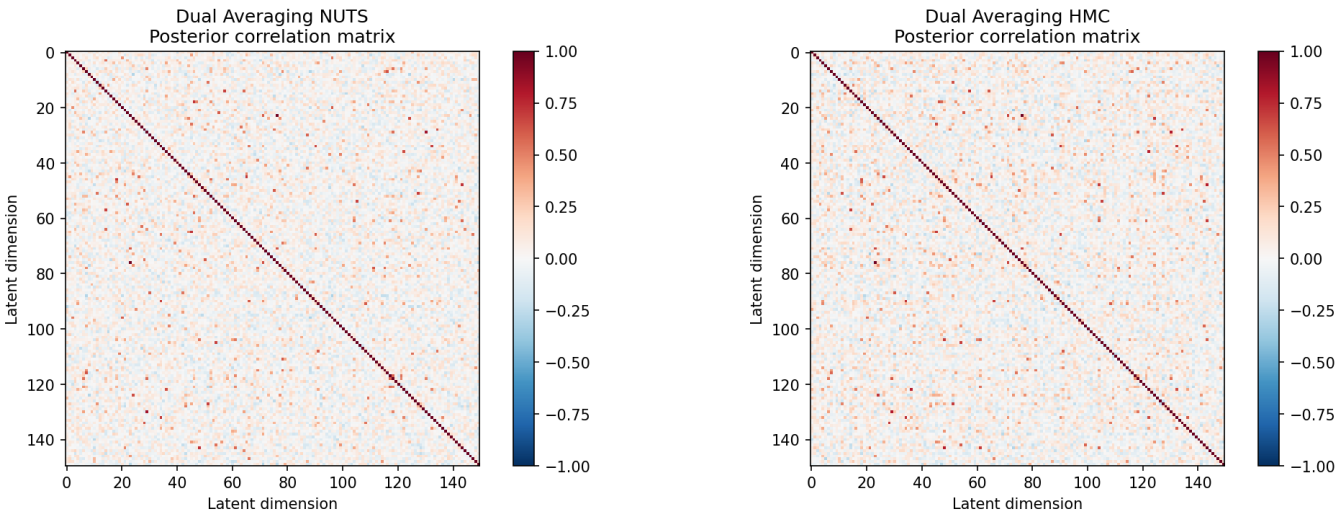


Figure 8: Posterior correlations among latent values f.

Appendix: GPC Efficiency

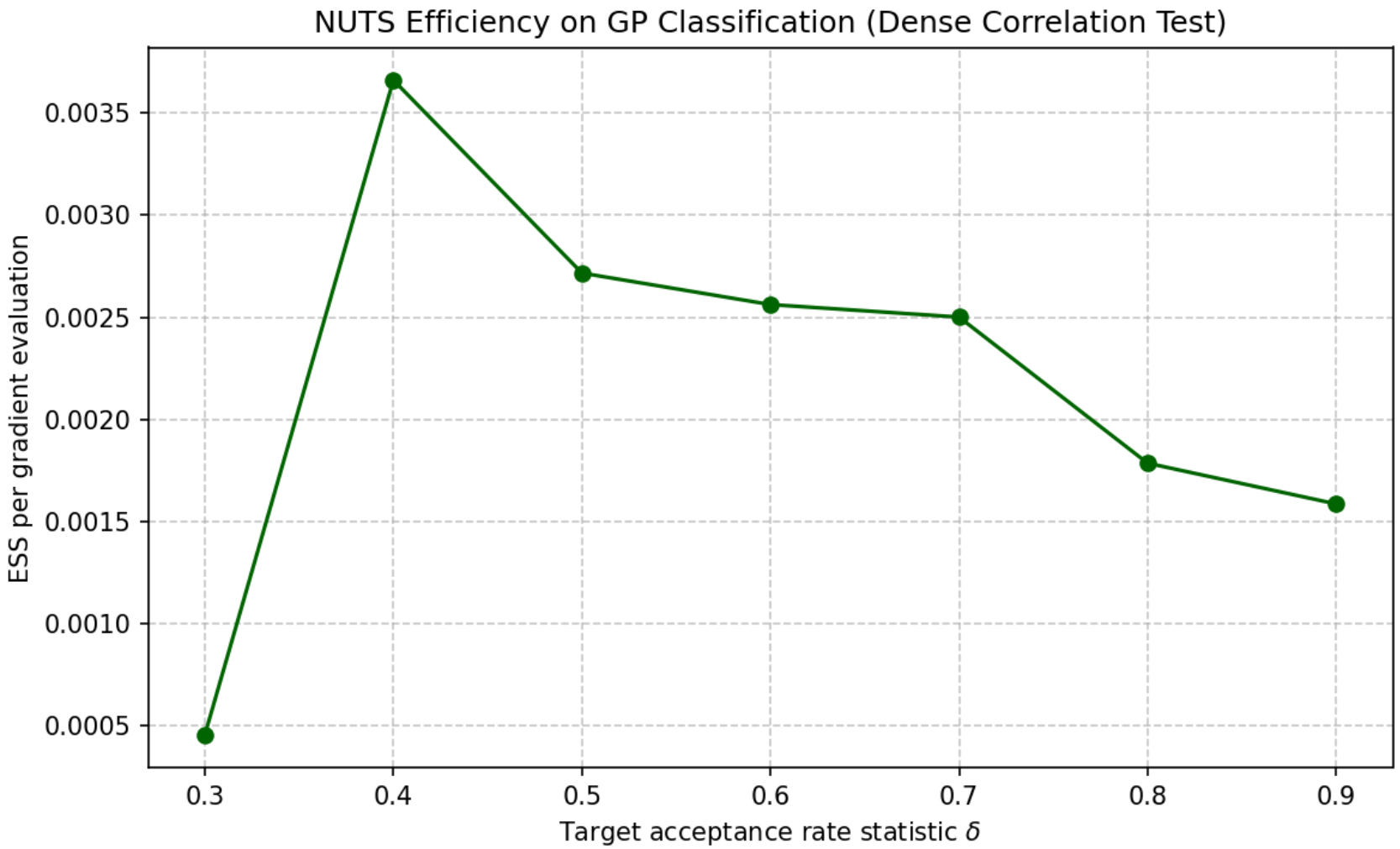


Figure 9: ESS/grad on GP classification. NUTS wins on per-gradient quality but at higher absolute cost.

Appendix: Funnel Scatter

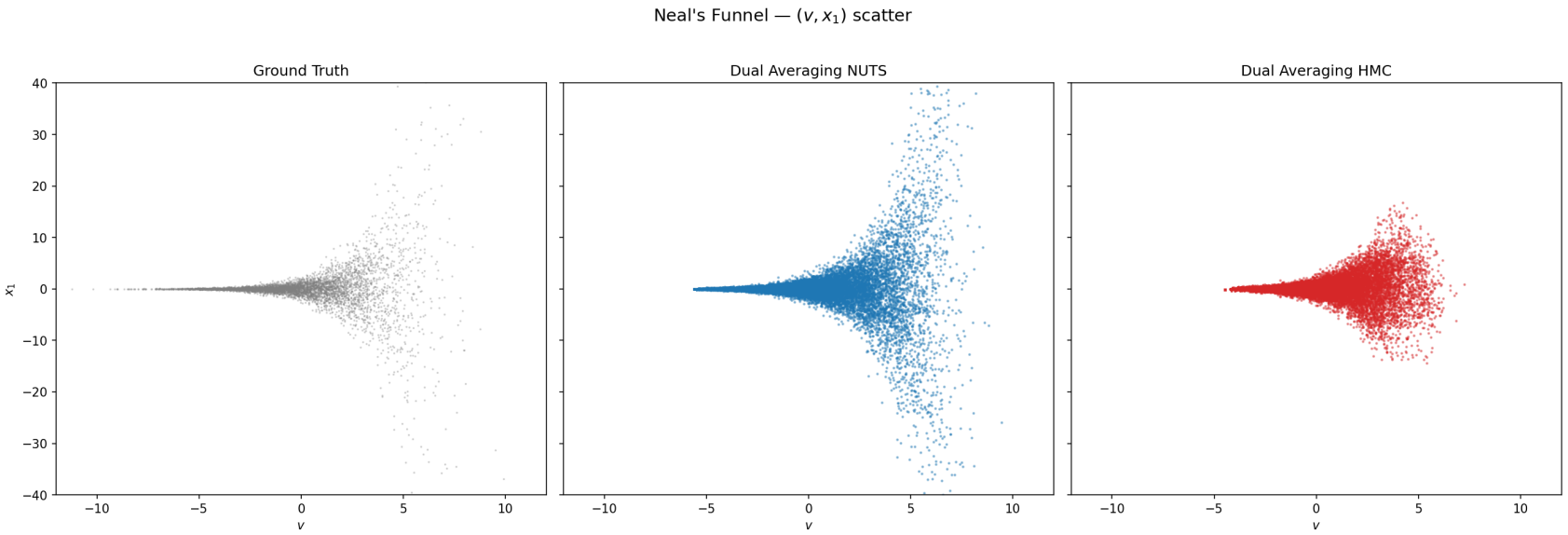


Figure 10: Left: ground truth. **Centre:** NUTS covers the full funnel. **Right:** HMC fails to explore it.

Appendix: Funnel Trace Plots & Marginal

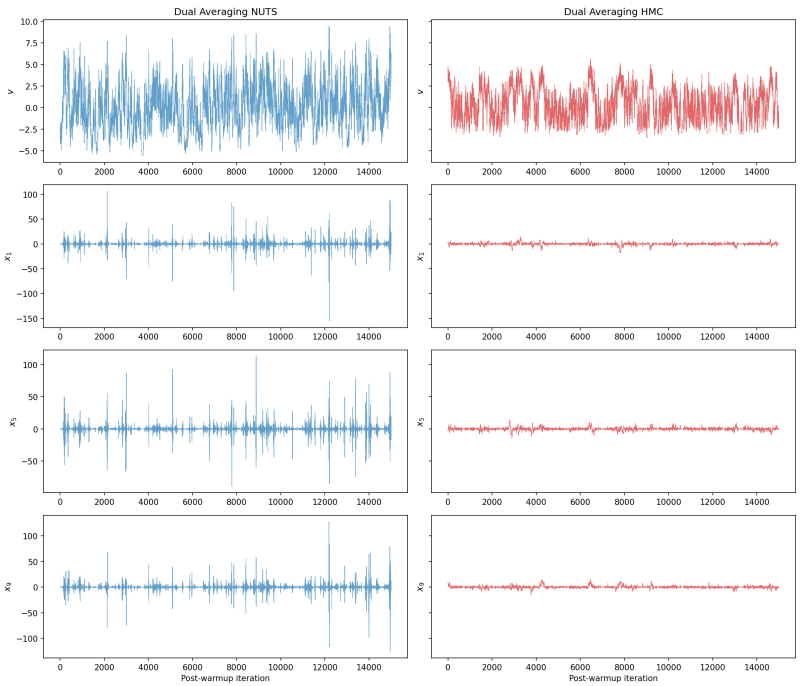


Figure 11: NUTS mixes freely; HMC shows flat segments.

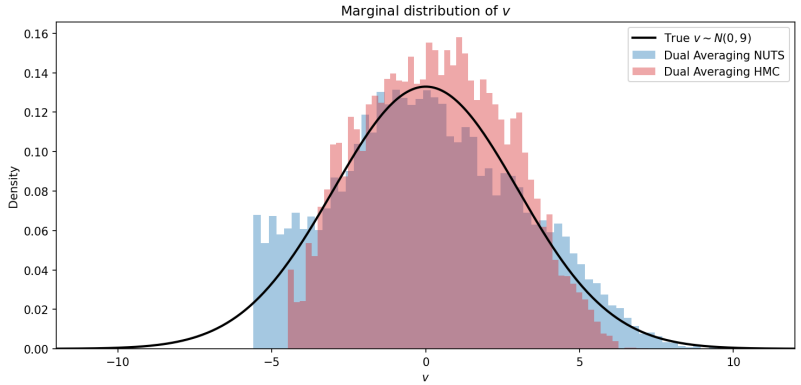


Figure 12: Marginal of v: NUTS recovers truth; HMC misses $v < 0$.

Appendix: Funnel Trajectory Lengths

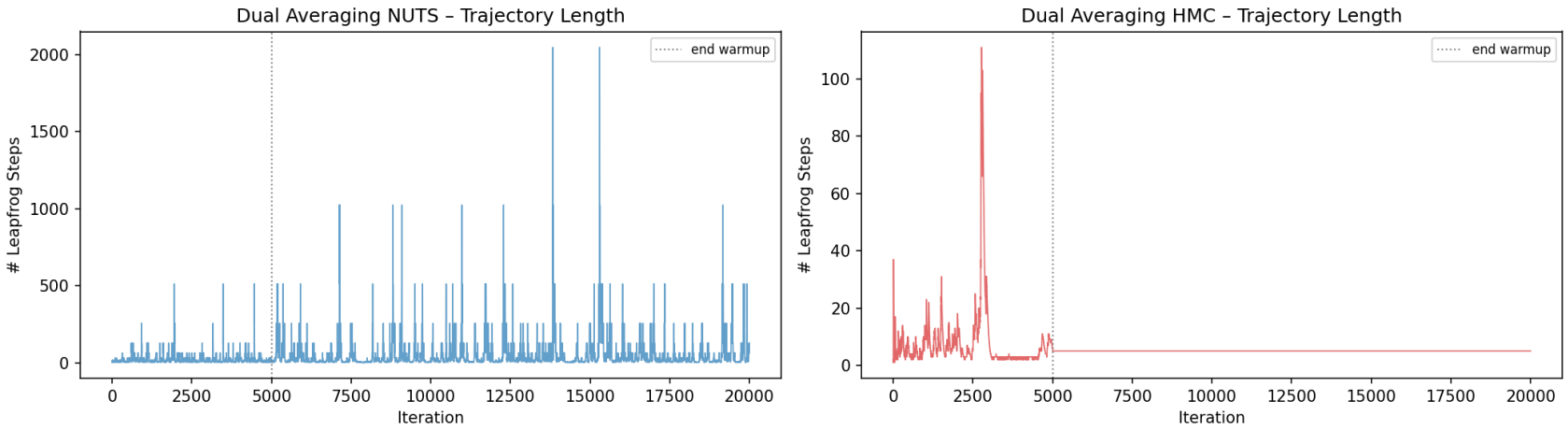


Figure 13: NUTS trajectory lengths on Neal's Funnel. Adaptive mechanism adjusts to local geometry.